

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_7049e1ed-b93\agent-af0a31b1d4ea5f7a8.jsonl`  
- SHA-256: `09f03e2ea8ff38d6dec6a6665890f13305b60f4b41b664ee068d74aeda4f7721`  
- Source modified: `2026-07-02T04:58:06+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_7049e1ed-b93`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T04:53:09.386Z)

You are an adversarial code reviewer. Today is 2026-07-02. Review PR #414 in the repo at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer (gh pr view 414, gh pr diff 414; the working copy is in worktree C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-claudemd-r25 ? read code there for context).
The PR claims to fix: R2-5 (CLAUDE.md 3 weeks false) (evidence in C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md ? Grep for the IDs; honor the verifier notes' scoping corrections).
Hunt REAL defects: correctness bugs in the diff, broken/vacuous tests, behavior changes beyond the brief, staging accidents (files that should not be in the diff ? check the file list!), convention violations (default-ON when brief says OFF, missing schema-version check, CWD assumptions), and whether the fix actually addresses the finding. READ-ONLY ? do not modify or comment on GitHub. blocker = must not merge; major = fix before merge; minor = nice-to-have.

2. user (2026-07-02T04:53:19.374Z)

```
title: docs(CLAUDE.md): truth-sync the stale Current-state section to shipped reality (audit R2-5)  
state: OPEN  
author: avidullu (Avi)  
labels:  
assignees:  
reviewers:  
projects:  
milestone:  
number: 414  
url: https://github.com/Khelsutra/badminton-highlight-indexer/pull/414  
additions: 20  
deletions: 13  
auto-merge: disabled  
--  
Fixes audit finding **R2-5 (P1)**: the agent-loaded `CLAUDE.md` "Current state" section was frozen at 2026-06-10 (62fb2fb) and had gone ~3 weeks false ? a fresh agent session reading it would conclude the platform/owned-model work never started and propose re-designing things that already shipped.
```

What / why

```
- **Rewrote "Current state" (now dated 2026-07-01)** against code + git log:  
  - Platform slice **shipped**: async job model (`backend/api/job_worker.py`; async  
    `/api/compile` #339), `COMPUTE_DECOUPLED_SERVING` M0?M9 largely shipped + M11 data-flywheel  
    (#336) + M12 gdrive-api (#338), SPA compute-picker (#335/#341), DB schema v9, packaging track in  
    progress.
```

- Owned-model track: `DECOUPLED\_COMPUTE` **delivered** on GCP (M0?M2 + M3.5, archived #270); Gen-0 weights `0.1.0-l4` pushed (n=2 plumbing-only); R4 density fix default-ON (#396).
- **Launch NO-GO recorded** (owner, 2026-06-30, #404) with its two unblockers (ship-gate target #172, WASB provenance).
- **Preserved the owner-gate language that is still true**, kept consistent with `docs/NEXT\_STEPS.md`'s NO-GO block: full-corpus training (n?5) + further owned-model milestones stay owner-gated; M7 persistent Cloud Run endpoint stays spend-gated; "do not start owner-gated work without explicit approval" retained.
- **Fixed the Gate-A claim** ("not wired into production"): Gate A is servable **opt-in** via `fusion\_hybrid`'s `min\_crossings` knob (default 0=OFF; served litmus neutral on production windowing, #401).
- **Flipped `StorageBackend`/`ComputeBackend` from "Future:" to shipped** (ABC + scheme factory; `backend/storage/`, `backend/compute/`) in "Architecture in one breath".

Docs-only; no code or behavior change. Every sentence was verified against the working tree / `git log` (PR numbers cited inline per the file's existing style).

Verification

- Ran locally: `python -m tools.check\_md\_links` ? clean (the `link-check` CI job's command).
- Verified each claim against the tree: `backend/api/job\_worker.py`, `backend/storage/` (local/gcs/gdrive/gdrive_api/s3 + `ref.py` `\_KNOWN` schemes), `backend/compute/base.py` + `cloud\_run.py`, `SCHEMA\_VERSION = 9` in `backend/infrastructure/database.py`, `fusion\_hybrid.py` + `min\_crossings` (default 0) in `backend/config/models.py`, and commits 8bcf6ce (#339), 334ad0f (#336), 819e395 (#338), 6d3d054/d92cc10 (#335/#341), 4134721 (#270), a555887, 4a7bfbfe (#404), c8a493d (#401).
- No Python touched, so no tests added; `run\_all\_tests.py` / ruff / mypy / coverage are unaffected and covered by CI.

Not a launch/bit-flip PR (no default flipped; no project reaches terminal state).

? Generated with [Claude Code](https://claude.com/claude-code)

====DIFF=====

diff --git a/CLAUDE.md b/CLAUDE.md

index d9807d9..5fe2e50 100644

--- a/CLAUDE.md

+++ b/CLAUDE.md

@@ -15,8 +15,9 @@ repos: `sports-data-collector`, a VLC-based `rally-annotator`.

(offline-segmenter narrative) and `docs/BACKLOG.md`.

- **Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation in progress):**

 `docs/HOW\_RALLY\_DETECTION\_WORKS.md` (2-layer mental model) ?

 `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md`

- (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built

 `rally\_gate.py`

- "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT\_HARNESS.md` (A/B + shadow

+ (fuse shuttle + person + audio cues; `rally\_gate.py` "Gate A" is now servable **opt-in** via the

+ `fusion\_hybrid` segmenter's `min\_crossings` knob, default 0=OFF ? the served litmus found it + neutral on production windowing, so NOT default; #401) ? `docs/EXPERIMENT\_HARNESS.md` (A/B + shadow

 eval so cues land flag-gated/default-OFF with zero regression) ?

 `docs/ROORA\_LABELING\_GUIDELINES.md`

 (v8 vendor labeling spec). **`docs/NEXT\_STEPS.md` has the prioritized pick-up for this track.**

- **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and

@@ -26,20 +27,26 @@ repos: `sports-data-collector`, a VLC-based `rally-annotator`.

 note ? update inbound refs ? keep the lineage ? then `git mv`, per the checklist + living lifecycle register in **`docs/DOC\_STATUS.md`**.

-## Current state (June 2026)

- **Scaffolding complete** ; strategy foundation + owned-model roadmap merged. **No

- platform/owned-model implementation has started. **A 2026-06-10 audit-driven hardening

- sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate, API

- validation, reliability, CI + test coverage) ? see git history.
- ****Two tracks, both owner-gated:**** the ****committed next PLATFORM slice = async job model****
- (``DEFERRED_HARDENING_PLAN.md`` #16, single-tenant); the ****owned-model track**** starts at a greenlit Phase-0 milestone (``OWNED_MODEL_IMPLEMENTATION_PLAN.md``). ****Do not start implementation without explicit owner approval.****
- +## Current state (2026-07-01)
- + ****Platform slice SHIPPED.**** Async job model is live (``backend/api/job_worker.py``; async ``/api/compile`` 202+poll, #339); ``COMPUTE_DECOUPLED_SERVING`` M0?M9 largely shipped + M11 data-flywheel (#336) + M12 gdrive-api (#338), default-OFF (ADR-014); SPA compute-picker (#335/#341); DB schema at v9. Packaging track (pip install ? ``indexer --app``, bundled SPA, torch-free) in progress ? ``docs/PACKAGING_PLAN.md``.
- + ****Owned-model / quality track:**** ``DECOUPLED_COMPUTE`` DELIVERED on GCP (M0?M2 + M3.5, archived #270); Gen-0 weights ``0.1.0-l4`` pushed (n=2 plumbing-only). R4 density fix promoted default-ON
- + (#396). ****Full-corpus training (n?5) + further owned-model milestones stay owner-gated**** (corpus
- + growth + greenlight per ``OWNED_MODEL_IMPLEMENTATION_PLAN.md``); the M7 persistent Cloud Run endpoint = owner spend-gated. ****Do not start owner-gated work without explicit approval.****
- + ****? Launch = NO-GO (owner, 2026-06-30, #404)**** ? we stay in development. Unblockers: ship-gate
- + target (#172, unset) + WASB weight provenance. Live priorities: ``docs/NEXT_STEPS.md`` (owner
- + priority = the UI-driven, alpha-shareable flow, item 0).
- ## Architecture in one breath
- ****Pluggable registries**** (ABC + ``Registry`` singleton): segmenters / providers / validators
- / sports / annotations ? see ``backend/pipeline/segmenters/base.py``. ****Future:****
- ``StorageBackend`` + ``ComputeBackend`` registries (PLATFORM §3).
- + / sports / annotations ? see ``backend/pipeline/segmenters/base.py``. ``StorageBackend`` + ``ComputeBackend`` are SHIPPED as ABC + scheme factory (``backend/storage/`` ?
- + local/gcs/gdrive/gdrive-api/s3 + ``ref.py``; ``backend/compute/`` ? Cloud Run) (PLATFORM §3).
- ``video_id`` = MD5 of the file ? ``backend/utls/hashing.py::compute_video_id`` (one helper).
- Typed config = ``backend/config/`` (Pydantic). DB schema evolves via the ``PRAGMA user_version`` migration ladder in ``backend/infrastructure/database.py``.

3. user (2026-07-02T04:53:19.505Z)

```

470-- verdict: CONFIRMED
471-[Omitted long context line]
472-[Omitted long context line]
473-
474-### R2-4 [P2/S] ? badminton-highlight-indexer
475-**No timeout on any serving-path ffmpeg subprocess ? one hung ffmpeg permanently wedges the single-worker job queue**
476-- verdict: CONFIRMED
477-[Omitted long context line]
478-[Omitted long context line]
479-
480-### R2-5 [P1/S] ? badminton-highlight-indexer
481-**Engine CLAUDE.md 'Current state' section is ~3 weeks false ? claims platform/owned-model work 'has not started'**
482-- verdict: CONFIRMED
483-[Omitted long context line]
484-[Omitted long context line]
485-
486-### R2-6 [P1/M] ? sports-data-collector
487-**Golden-corpus registry rot: all 6 owned 'curated' videos point at a deleted absolute local_path and the 4.9GB source videos are single-copy with no documented backup**
488-- verdict: CONFIRMED
489-[Omitted long context line]
490-[Omitted long context line]

```

4. user (2026-07-02T04:53:30.279Z)

Correctly-scoped fix (S): (a) don't start the 333 ms clock until the panel is first shown and pause/clear it when hidden ? gate it inside ``toggle()`` in `panel.ts`; (b) make render's clock read

non-scanning (e.g. a `peekNow()` on `DirectVideoHandler` that uses only the cached element), keeping the full scan for explicit triggers (Refresh button, panel open). Optional follow-up (M, separate decision): replace the `<all_urls>` content script with `activeTab + scripting.executeScript` on icon click, which eliminates injection on every page entirely and drops the broad host permission ? aligns with LD-7's own rationale, worth raising but not required to fix the perf debt. Not OWNER_GATED ? pure code change, no spend or product pivot.

R2-2 [P2/S] ? rally-annotator

****Locked decision LD-i8 (web language persisted + seeded from browser language) is documented as shipped but not implemented ? panel always boots in English and forgets the choice on reload****

- verdict: CONFIRMED

- reasoning: Doc claims shipped: `rally-annotator/docs/LOCALIZATION.md:55-57` (LD-i8: "Web: persisted in `chrome.storage.local` (seeded from the browser language; default en)") and :99-100 mark the web i18n phases done (PR #18 "selector", PR #19); `web/README.md:22-23` says "persisted in-panel language selector". Code says otherwise: (1) `web/src/i18n/index.ts:56` boots the module-level locale to `DEFAULT_LOCALE ("en")` ? a fresh content-script injection always starts English; (2) `negotiateLocale (index.ts:72-80)`, which implements the stored?navigator.languages?en order, has ZERO runtime callers ? grep across web/ shows its only callers are unit tests (`web/test/i18n.test.ts:47-52`); (3) `panel.ts:18` declares `onLocaleChange` with the comment "optional: persist the choice (content script wires this)", but the actual `mountPanel` call in `web/entrypoints/content.ts:32-42` passes only `annotator/video/identity/download ? onLocaleChange` is never wired, so the change handler at `panel.ts:306-311` updates only the in-memory `current`; (4) no storage read/write of any language key exists anywhere: `persist/store.ts` handles only "rally:"-prefixed rows, `background.ts` (all 27 lines) handles only downloads and icon clicks. Git: repo is at origin/main (3d604bc); no commit after 0a05ecf (#19) touches web locale persistence ? 3d604bc ("docs: sync READMEs, multi-agent audit fixes") actually re-affirmed the false "persisted" README claim. The E2E (`web/e2e/extension.spec.ts:80-101`) switches the locale within one page session and never reloads, so it can't catch this. Failure scenario: a Hindi rater (browser hi-IN) opens a video ? panel boots in English despite the "seeded from browser language" claim; they select ??????; on the next page load/navigation the panel is English again and the choice must be re-made on every video.

- notes: Claim verified exactly as stated; not OWNER_GATED (LD-i8 is already the locked decision ? this is pure implementation). Sharper framing: both halves of LD-i8-web are missing (no boot-time seeding AND no persistence), yet all plumbing already exists and is tested, which is why the fix is S: in `content.ts` main(), read e.g. `lang` from `browser.storage.local`, call `setLocale(negotiateLocale(stored, navigator.languages))` before `mountPanel`, and pass `onLocaleChange: (l) => void browser.storage.local.set({ lang: l });` add a unit test for the wiring and extend the Playwright E2E with a reload-persists-locale assertion (the current per-locale E2E never reloads, which is the coverage gap that let the docs drift). Scope note: the VLC half of LD-i8 (config-file persistence, PRs #20/#21) was not part of this claim and was not verified either way. Severity P2: no data loss/crash, but a locked decision is documented and README-advertised as shipped while absent ? it defeats the localization value prop for the target hi/kn/te raters (English boot every page load) and the doc-sync commit 3d604bc propagated the false claim.

R2-3 [P2/S] ? sports-data-collector

****Collector pins obsreport as a plain PyPI requirement while the name is unclaimed on PyPI ? dependency-confusion window and install broken today****

- verdict: CONFIRMED

- reasoning: (1) Bare PyPI-style pin exists in current tree: `C:/Users/avidu/Projects/khelsutra-guru/sports-data-collector/pyproject.toml:38-40` ? `report = ["obsreport>=0.1.3"]`, with the comment at lines 32-37 itself admitting the intended form is a git+ssh direct reference. (2) Name is unclaimed on PyPI today (2026-07-02): `https://pypi.org/pypi/obsreport/json` returns 404, and `pip index versions obsreport` returns "ERROR: No matching distribution found for obsreport" (personally observed). (3) The private package really is named `obsreport` (`sports-obsreport/pyproject.toml:6`, version 0.1.3, tag v0.1.3 exists) and lives at the private repo `Khelsutra/sports-obsreport` (gh api confirms private=true; `avidullu/sports-obsreport` redirects there). (4) Contrast confirmed: the engine already pins it safely ? `badminton-highlight-indexer/pyproject.toml:68` `obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3`. (5) No fix landed: the bare pin was introduced in commit 038424d and no later commit touched it (git log -S on `pyproject.toml`); HEAD adcd206, working tree clean. Failure scenario A (today): `pip install .[report]` in any clean env fails resolution ? the extra is uninstallable unless the sibling checkout was pre-installed editable. Failure scenario B (window): anyone can register `obsreport` on PyPI and publish `>=0.1.3`; a later `pip install .[report]` then silently executes attacker code (`sdist build hooks/import`) instead of failing

loudly.

- notes: Severity calibrated down from an implied P1: the exposure is real but narrow. `report` is an optional extra; the base install is unaffected; the collector treats obsreport as a soft dependency that no-ops when absent (sports-data-collector/src/reporting/__init__.py:15-22); and CI installs only `[dev]` (.github/workflows CI line 19), so no automated pipeline ever resolves the bare name ? the confusion window only fires when a human/agent runs `pip install .[report]` in a fresh env (in the documented sibling-editable workflow, pip sees obsreport 0.1.3 already satisfied and never queries PyPI). It is still a standing security window on a generic, squatable name, plus a broken-today extra, in a repo whose sibling project already solved the identical problem. Fix (S, one line): replace pyproject.toml:39 with `obsreport @ git+ssh://git@github.com/Khelsutra/sports-obsreport.git@v0.1.3` (prefer the canonical Khelsutra org path ? the avidullu path in the engine's pin only works via GitHub's transfer redirect and is itself mildly stale; consider updating badminton-highlight-indexer/pyproject.toml:68 too). setuptools accepts PEP 508 direct references without extra config (no hatchling-style allow-direct-references needed), and since sports-data-collector is not published to PyPI, the PyPI ban on direct references does not apply. Optional hardening (owner-gated, not required for the fix): claim the `obsreport` name on PyPI with a stub to close the squatting window permanently.

R2-4 [P2/S] ? badminton-highlight-indexer

****No timeout on any serving-path ffmpeg subprocess ? one hung ffmpeg permanently wedges the single-worker job queue****

- verdict: CONFIRMED

- reasoning: Single-worker executor: backend/api/job_worker.py:47-49 (max_workers=1) with the in-code "O2 RISK" admission at :32-39 that one hung job blocks all subsequent jobs and "the executor itself cannot enforce timeouts on the Thread". Serving-path ffmpeg/ffprobe calls with NO timeout=: (compile job, POST /api/compile main.py:1594 -> _execute_compile -> compiler.compile_highlights at main.py:1686, no wrapper) backend/pipeline/stitcher/compiler.py:130 (ffprobe), :321 (per-clip libx264 encode), :494 (concat); (process job, _execute_processing main.py:1132) validators backend/pipeline/validators/format.py:43, pts_continuity.py:35, resolution_fps.py:15; backend/utls/video.py:18,:75 (chunking); backend/pipeline/segmenters/gemini.py:166; backend/pipeline/audio/hit_detector.py:53. The mitigations listed in job_worker.py:34-37 do NOT cover ffmpeg: run_bounded wraps only the Gemini generate call (backend/providers/gemini.py:233, op_timeout_sec=180); detector timeout_sec covers only GPU runner subprocesses (detectors/base.py:116, native_wasb_runner.py:134). Recovery is restart-only: fail_stale_jobs is called once at startup (main.py:2030; database.py:1122-1125) and marks ALL queued/running jobs failed. No fix landed: git head e410136; commit 24178a7 (#374) documents the O2 risk but adds no timeouts. Secondary confirmed defect: /api/source proxy warm (main.py:657 daemon thread) runs annotation_proxy.py:81 ffmpeg with no timeout; a hang leaks the _proxy_inflight dedup entry forever (main.py:607-620), permanently disabling proxy generation for that video (does not wedge the job queue).

- notes: Claim is literally true, but "permanently wedges" should be "wedges until server restart, and the restart sweep (fail_stale_jobs) then fails every queued job". Sharper failure scenario: POST /api/compile for a video whose stored local filepath sits on a network-backed mount (Google Drive streaming / SMB) ? storage.acquire_local_input is identity for local paths (main.py:1683) ? the mount stalls mid-read (or the MP4 is truncated/corrupt and ffmpeg's demuxer loops); subprocess.run at compiler.py:321 blocks forever; the compile job stays 'running'; every subsequent /api/process and /api/compile returns 202 but stays 'queued' indefinitely with zero error surfaced (drains submitted at main.py:1635 queue behind the blocked task on the max_workers=1 executor); only a full restart recovers, at the cost of failing the whole queue. Severity P2 not P0/P1: the trigger class is rare (mitigations already bound the historically hang-prone ops ? Gemini generate 180s, GPU/WSL detector timeout_sec 1800s, Cloud Run poll max_wait_sec), the risk is explicitly documented in-code, and restart recovers; but impact when triggered is a silent total outage of the job pipeline. Fix (S, mechanical): add timeout= to the ~10 serving-path ffmpeg/ffprobe subprocess.run/check_output sites (config-driven, e.g. 30-60s for ffprobe probes, stitching-level 1800s default for encode/concat/chunking) and map subprocess.TimeoutExpired to the existing clean-job-failure paths (compiler already converts CalledProcessError to a failed-job result at main.py:1687-1695; annotation_proxy already catches broadly at annotation_proxy.py:82). The broader per-job wall-clock timeout job_worker.py:38 calls a "future slice" is a separate M item and not required to close this finding. Not OWNER_GATED ? timeout defaults are an engineering choice with no spend/product implications.

R2-5 [P1/S] ? badminton-highlight-indexer

****Engine CLAUDE.md 'Current state' section is ~3 weeks false ? claims platform/owned-model work 'has not started'****

- verdict: CONFIRMED

- reasoning: Personally reproduced against the working tree and git history. (1) C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\CLAUDE.md:30-31 states "***No platform/owned-model implementation has started.**"; lines 34-37 say the "committed next PLATFORM slice = async job model" and the owned-model track "starts at a greenlit Phase-0 milestone... Do not start implementation without explicit owner approval"; lines 41-42 call `StorageBackend` + `ComputeBackend` registries `Future:*. git blame shows lines 30-37 were last written 2026-06-10 (commit 62fb2fb) and never touched since, despite CLAUDE.md itself being edited 6+ times after (e735fd4 06-13 ... af1ccbe 06-30 ? all other sections). (2) Current-tree reality contradicts every one of those claims: backend/storage/ contains a StorageBackend ABC + local/gcs/gdrive/gdrive_api/s3/capabilities implementations; backend/compute/base.py + cloud_run.py exist (ComputeBackend); backend/api/job_worker.py + tests/test_async_jobs.py exist ? the "committed next slice" async job model already shipped (8bcf6ce #339 "make /api/compile async (202 + poll)", 2026-06-22); backend/flywheel.py + backend/worker + backend/eval/served_gate_a.py exist. Owned-model track: a555887 (2026-06-20) "M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed"; 4134721 archives DECOUPLED_COMPUTE as "DELIVERED (M0-M2 + M3.5 on GCP)"; 334ad0f (#336) M11 data-flywheel; b72e6de (#319) training worker fix. 314 commits landed since 2026-06-13, through 2026-07-01 (e410136), including a recorded launch NO-GO decision 2026-06-30 (4a7bfbe #404) ? i.e., the project reached launch-consideration, the opposite of "not started". (3) No fix landed: git log -- CLAUDE.md shows the last commit touching it (af1ccbe, 2026-06-30) added only the bit-flip-PR convention + Python 3.10 line; the Current state section is untouched since 06-10. Bonus staleness in the same file: lines 18-19 claim rally_gate.py Gate A is "not wired into production" while backend/eval/served_gate_a.py and c8a493d "served Gate-A litmus" (#401) indicate a served Gate-A path now exists.

- notes: Sharper failure scenario: CLAUDE.md is auto-loaded project memory for every fresh agent session in this heavily agent-driven repo. A fresh session reads "No platform/owned-model implementation has started" + "Do not start implementation without explicit owner approval" and will (a) tell the owner the async job model is an un-started, gated future slice and propose designing/implementing what already exists (backend/api/job_worker.py), (b) treat StorageBackend/ComputeBackend as "Future" and re-design duplicates of backend/storage/ and backend/compute/, (c) misreport project status upward (e.g., in handoffs/audits) as pre-implementation when Gen-0 weights shipped and a launch NO-GO was formally recorded 06-30. This also directly violates the workspace's own "keep it honest ? reflects real, shipped state" convention. Correctly-scoped fix (S, one doc-only PR): rewrite CLAUDE.md "Current state" (lines 29-37) to reflect reality as of 2026-07-01 ? platform slice (async jobs, storage/compute backends, capabilities/assets endpoints) SHIPPED; owned-model M0-M2+M3.5 DELIVERED on GCP, Gen-0 weights pushed; launch NO-GO 2026-06-30 with pointer to #404/docs; fix line 41-42 ("Future" ? "implemented, see backend/storage/, backend/compute/"); fix lines 18-19 Gate-A "not wired" claim against backend/eval/served_gate_a.py; re-point "committed next slice" at whatever docs/NEXT_STEPS.md currently prioritizes. Not OWNER_GATED ? this is a factual truth-sync of docs to shipped code, the same class as the repo's prior "audit truth-sync" commits (62fb2fb, 9736a3b); no spend or strategy decision required. Round-1 claim of "~3 weeks false" is accurate: section frozen 2026-06-10, contradicting implementation landed from ~2026-06-20 onward.

R2-6 [P1/M] ? sports-data-collector

Golden-corpus registry rot: all 6 owned 'curated' videos point at a deleted absolute local_path and the 4.9GB source videos are single-copy with no documented backup

- verdict: CONFIRMED

- reasoning: Registry rot half CONFIRMED by direct observation. sports-data-collector/data/sports_metadata.db (git-tracked; rows landed in commit 9e6f204 "#19 ingest the n=6 owned golden corpus", DB last touched 9f1e792 with no path fix since): videos table has 8 status='curated' rows; the 6 owned ones (adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles; license_type=Proprietary, is_commercial=1) all carry local_path under 'C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled*.MP4' ? that directory does not exist. The folder moved to 'C:\Users\avidu\Projects\khelsutra-guru\Annotation Setup\' whose 'Golden Labelled\' now contains ONLY the six .rallies.csv label files; zero copies of the MP4s exist anywhere in the workspace (searched). The committed export artifact data/eval_export/manifest.json repeats the same 6 dead absolute paths. Code consequence at sports-data-collector/src/cli/curate.py:679-684: export of a curated video with missing local_path emits md5=None plus a warning, and src/core/manifests.py:23-26 documents md5 as "the indexer's join key" ? so any fresh export of the entire owned golden corpus loses the deterministic MD5 join to badminton-highlight-indexer. Data-loss half CONFIRMED as single-copy but the "no documented backup" wording is wrong: (a) verified live via read-only 'gcloud storage ls' (2026-07-02): gs://khelsutra-rally-corpus/videos/ has 10 objects and NONE of the six originals, while labels/ DOES hold all six

(2026-06-08_adarsh_avi_singles.rallies.csv etc.) and trajectories are in the bucket ? annotation work-product is safe, source footage is not; (b) Google Drive golden folder (G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15) holds only the 7 proxy clips, not the original six; (c) the sole remaining copy is documented at F:\[Khelsutra] GoPro Hero Black 12\KheISutra\Training\Golden Labelled\ per badminton-highlight-indexer/scratch/copy_videos_to_gcs.ps1:31-36 and docs/DATA_IN_GCS.md:153 (\$6 Phase-2) ? and the F: drive is NOT currently mounted (only C: and G: exist), so even that copy's integrity is unverifiable today, and no MD5s of the six were ever persisted anywhere (eval_export/manifest.json predates the md5 field from PR #51; no proxy-registry rows).

- notes: Corrections to the round-1 claim: (1) Size is ~37.9 GB, not 4.9 GB ? per-clip sizes annotated in scratch/copy_videos_to_gcs.ps1:31-36 (10.9+7.7+11+6.7+1.4+0.19 GB); the ~4.9-5.2 GB figure matches the three unrelated Roora pilot videos under 'Annotation Setup/Archive/.../Annotation Videos/'. (2) "No documented backup" is false ? badminton-highlight-indexer/docs/DATA_IN_GCS.md \$5/\$6/\$8 explicitly documents the F: archive location and a Phase-2 migration plan (tracker at doc line 246: 14/15 videos still unmigrated); but "single-copy" stands: exactly one copy of the six source videos exists, on an intermittently-mounted external HDD, unverifiable right now, with no persisted hashes. (3) Downgrade from potential P0 to P1 because labels + trajectories for all six ARE in gs://khelsutra-rally-corpus (verified), so GPU-free eval/litmus and synth-train survive a drive failure; what would be permanently lost is the proprietary source footage (no re-extraction via --trajectory-source detector, no re-annotation, no re-rendering). Sharper failure scenarios: (a) F: HDD failure = permanent loss of ~38 GB of irreplaceable owned golden footage; (b) today, 'curate export' on the owned corpus emits md5=None for all six (curate.py:679-684), silently breaking the indexer MD5 join contract added in PR #51. Correctly-scoped fix (M, cross-repo): mount F:, verify + hash the six MP4s (persist MD5s into the DB/manifest immediately ? S and decision-free), generate 1080p proxies, upload via the existing scratch/copy_videos_to_gcs.ps1 or the rclone cloud path (Phase-2, already planned), then reimport/annotate local_path (or add a gs:// video_uri column) in sports_metadata.db and refresh data/eval_export/manifest.json. Partially OWNER_GATED sub-decision only: DATA_IN_GCS.md \$7.4 leaves proxies-only vs full-4K migration scope OPEN and the upload is real (though tiny, cents/month) GCS spend; the hash-and-record step and the DB path fix need no owner decision.

R2-7 [P2/S] ? non-git

****Roora vendor pilot record (paid deliverables + corrected golden labels + 3 videos) is a single unversioned local copy at risk of loss****

- verdict: CONFIRMED

- reasoning: CONFIRMED for the paid deliverables + corrected golden labels; the "3 videos" component of the claim is REFUTED (Drive copies exist). Evidence: (1) Single-copy artifacts verified on disk: raw paid CVAT deliverables at 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup/Archive/Roora Pilot Review - 6 June 2026/' ? badminton/{badminton, badminton (1), badminton (2)}/Annotations/*.xml (~2.3 MB per-frame boxes), pickleball/*.zip (3 zips, 260 KB), tabletennis/*.zip (3 zips, 213 KB) ? plus the human-corrected golden labels 'Annotation Setup/golden_labels_badminton.csv' (2.5 KB, mtime 2026-06-06) and the rescued multisport goldens 'Annotation Setup/Trajectories/roora/{pickleball,tabletennis}_rallies.csv'. (2) Not in git: sports-data-collector tracks only the AS-DELIVERED parse (data/roora_badminton_canonical.csv, verified via git ls-files + diff ? it still has rally 2 = 0:02.503?0:32.533, rally 7 end 1:51.612, no rallies 7.5/16.5), while golden_labels_badminton.csv holds the corrections (rally 2 start?0:29, rally 7 end?1:38, missed rallies 7.5@1:46 and 16.5@3:34 added, winner?out fixes) ? those corrections exist NOWHERE else (find over C:/Users/avidu/Projects + Downloads returned only this one file). (3) Not in GCS: recursive 'gcloud storage ls -r gs://khelsutra-rally-corpus/**' contains zero objects matching roora|pilot|pickle|tabletennis|sample_long|golden_labels; labels/ holds exactly the 15 canonical corpus clips (pilot video not among them); archive/ holds only a droplet archive. (4) Not on Drive: 'G:/My Drive/Sharing' has guidelines/briefs and 'Golden Annotations' (MahadevpuraSingles only) ? no CVAT exports. (5) Outside every backup convention: badminton-highlight-indexer/docs/DATA_IN_GCS.md \$5 gap table (lines 117?131) covers only the 15-clip corpus and never mentions the pilot artifacts (and its line 121/142 path 'C:\?Annotation Setup\Collect\Trajectories\' is stale ? the folder moved, no Collect dir exists); the F: backup (memory checkpoint 2026-06-22) covered only the 7 new-clip labels; docs/archives/roora-vendor/README.md archives only guidelines+ADR, not pilot results; F: is currently unmounted. REFUTED part: the 3 pilot videos have byte-size-identical cloud copies at 'G:/My Drive/Sharing/Annotation Videos/Pilot Videos/' (Sample Long Video - Baadminton.MP4 = 2,882,861,900 B; Video 2 - Pickleball.mp4 = 1,435,276,306 B; Video 3 - Tabletennis.mp4 = 871,944,715 B ? all match local), and the local 'Annotation Videos/desktop.ini' references GoogleDriveFS.exe proving Drive File Stream provenance. Owner's own memory audit had flagged the

precursor ("Roora multisport CSVs live ONLY in %TEMP%? one cleanup from loss"); the rescue moved them to Annotation Setup but they remain a single unversioned C:-disk copy.

- notes: Correction to the finding: the at-risk payload is NOT the 4.9 GB folder ? the videos are safe on the owner's Google Drive. What is genuinely one C:-disk failure (or one folder-cleanup) from loss is ~3 MB of paid, partly irreplaceable work product: the raw vendor CVAT exports for all 3 sports (box geometry the tracked canonical CSVs discard), the human frame-by-frame correction record (golden_labels_badminton.csv + rally_review.csv ? hours of review labor and the calibration evidence behind the vendor feedback), the pickleball/tabletennis golden CSVs backing the recorded TT-0.909/PB-0.308 multisport result, and the pilot review MD/PDF record. Vendor re-export from CVAT job 4075945 is not under owner control. Failure scenario: C: SSD failure or a cleanup of the non-git 'Annotation Setup' folder silently destroys the only record of the paid pilot and its corrected labels; nothing in DATA_IN_GCS.md's migration plan would ever pick it up because the pilot artifacts are absent from its \$5 gap table. Fix (S, not owner-gated for the metadata): one 'gcloud storage cp -r' of 'Annotation Setup/Archive/Roora Pilot Review - 6 June 2026/' (excluding Annotation Videos/, already on Drive) + 'Annotation Setup/{golden_labels_badminton.csv,rally_review.csv,Trajectories/roora/}' to gs://khelsutra-rally-corpus/archive/roora-pilot-2026-06/ (archive/ prefix precedent already exists; ~3?10 MB, negligible cost), then add a Roora-pilot row to DATA_IN_GCS.md \$5 and fix its stale 'Annotation Setup\Collect\Trajectories' path (lines 121/142). Including the 3 pilot videos in the bucket is the only owner-gated piece (DATA_IN_GCS \$7 open decision 4, ~5 GB ? \$0.10/mo) and is optional given the Drive copies.

R2-8 [P1/S] ? badminton-highlight-indexer

****Upload retention sweep tool exists but nothing schedules it ? unbounded upload growth + unenforced 7-day privacy retention on any hosted deployment (#301)****

- verdict: CONFIRMED

5. user (2026-07-02T04:53:31.341Z)

```
256:[Omitted long matching line]
480:### R2-5 [P1/S] ? badminton-highlight-indexer
481:**Engine CLAUDE.md 'Current state' section is ~3 weeks false ? claims platform/owned-model
work 'has not started'**
483:[Omitted long matching line]
484:[Omitted long matching line]
556:- [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra,
sports-data-collector, rally-annotator, sports-obsreport: Engine CLAUDE.md 'Current state' is ~3
weeks false ? claims platform/owned-model work 'has not started'
601:- khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-
data-collector, rally-annotator, sports-obsreport: Agent/contributor conventions exist only in
the engine: CLAUDE.md and PR templates missing or inconsistent elsewhere (hygiene, `badminton-
highlight-indexer/CLAUDE.md`)
```

6. user (2026-07-02T04:53:50.787Z)

```
CLAUDE.md
---
{"base":"master","commits":["3b36f9f1a268688c124f76f2c9733043cd8dd433"],"head":"docs/claude-md-
current-state-truth-sync","mergeable":"MERGEABLE"}
```

7. user (2026-07-02T04:53:50.810Z)

```
1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
9        contracts (the cold-start map; read before touching code).
10     - **Authoritative owned-model roadmap** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+
the
11     live `docs/NEXT_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where
```

```

they
12     disagree (licensing, Gemma-4 stance, base model, conversational-QA).
13     - **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`,
`docs/PMF_MARKET_RESEARCH.md`,
14     `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also
`docs/ROADMAP.md`
15     (offline-segmenter narrative) and `docs/BACKLOG.md`.
16     - **Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation
in progress):**
17     `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ?
`docs/MULTI_SIGNAL_FUSION_PLAN.md`
18     (fuse shuttle + person + audio cues; `rally_gate.py` "Gate A" is now servable **opt-
in** via the
19     `fusion_hybrid` segmenter's `min_crossings` knob, default 0=OFF ? the served litmus
found it
20     neutral on production windowing, so NOT default; #401) ? `docs/EXPERIMENT_HARNESS.md`
(A/B + shadow
21     eval so cues land flag-gated/default-OFF with zero regression) ?
`docs/ROORA_LABELING_GUIDELINES.md`
22     (v8 vendor labeling spec). **`docs/NEXT_STEPS.md` has the prioritized pick-up for this
track.**
23     - **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
24     **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). **Archive policy:** only
move
25     **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level ? and
**before
26     archiving, HONESTLY update the doc first** (verify claims vs code ? flip status + a
completion
27     note ? update inbound refs ? keep the lineage ? then `git mv`), per the checklist +
living
28     lifecycle register in **`docs/DOC_STATUS.md`**.
29
30     ## Current state (2026-07-01)
31     - **Platform slice SHIPPED.** Async job model is live (`backend/api/job_worker.py`;
async
32     `/api/compile` 202+poll, #339); `COMPUTE_DECOUPLED_SERVING` M0?M9 largely shipped +
M11
33     data-flywheel (#336) + M12 gdrive-api (#338), default-OFF (ADR-014); SPA compute-
picker
34     (#335/#341); DB schema at v9. Packaging track (pip install ? `indexer --app`, bundled
SPA,
35     torch-free) in progress ? `docs/PACKAGING_PLAN.md`.
36     - **Owned-model / quality track:** `DECOUPLED_COMPUTE` DELIVERED on GCP (M0?M2 + M3.5,
archived
37     #270); Gen-0 weights `0.1.0-l4` pushed (n=2 plumbing-only). R4 density fix promoted
default-ON
38     (#396). **Full-corpus training (n?5) + further owned-model milestones stay owner-
gated** (corpus
39     growth + greenlight per `OWNED_MODEL_IMPLEMENTATION_PLAN.md`); the M7 persistent Cloud
Run
40     endpoint = owner spend-gated. **Do not start owner-gated work without explicit
approval.**
41     - **? Launch = NO-GO (owner, 2026-06-30, #404)** ? we stay in development. Unblockers:
ship-gate
42     target (#172, unset) + WASB weight provenance. Live priorities: `docs/NEXT_STEPS.md`
(owner
43     priority = the UI-driven, alpha-shareable flow, item 0).
44
45     ## Architecture in one breath
46     - **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers /
validators
47     / sports / annotations ? see `backend/pipeline/segmenters/base.py`. `StorageBackend` +
48     `ComputeBackend` are SHIPPED as ABC + scheme factory (`backend/storage/` ?
49     local/gcs/gdrive/gdrive-api/s3 + `ref.py`; `backend/compute/` ? Cloud Run) (PLATFORM
S3).

```

```

50     - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
51     - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
52       user_version` migration ladder in `backend/infrastructure/database.py`.
53
54     ## Committed guardrails (non-negotiable)
55     - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
56       training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
57       code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-
agreed).
58     - **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
**Gemma-4
59       is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
60       needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license).
**Exclude
61       minors** from the training pool; per-user training data needs a separate explicit opt-
in.
62     - **v1 = rally + highlight + history + correction loop** ; **gait/biometrics strictly
future**
63       (GDPR Art. 9).
64     - **Generic data schema** ; `coach ? athlete` is an **optional overlay** , not baked in.
65     - **Compute = GCP ONLY (ADR-013, 2026-06-20):** GPU training + serving run on **GCP**
(`deploy/gcp/` ;
66       Cloud Run+GPU for serving per ADR-012). **DigitalOcean is DROPPED for compute** (DO
GPU not enabled
67       on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't
re-evaluate
68       DO/Paperspace for compute without a new ADR; the `$200` DO credits go to **non-compute
only** . The
69       `deploy/digitalocean/` scripts are kept solely as provider-agnostic Linux-GPU setup
reused by GCP.
70
71     ## Workflow conventions
72     - **Branch off `master` ; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
73     - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers`
package requires ?3.10); `ffmpeg` required.
74     - Never push to the default branch directly; descriptive commit messages.
75     - **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` $7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
76     - **Launch / "bit-flip" PRs** (a default-flip that drives a tracked project to terminal
state) MUST, in the SAME PR, run the `docs/DOC_STATUS.md` $2 archival due-diligence on that
project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR
lineage ? `git mv` to `docs/archives/` ? record in DOC_STATUS $3), and fix any internal-link
breakage the move causes. Enforced by `.github/PULL_REQUEST_TEMPLATE.md` + the `link-check` CI
job (`python -m tools.check_md_links`).
77     - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to
the owner,
78       who has the golden corpus + labelled videos; verify pure-logic paths locally.
79

```

8. user (2026-07-02T04:54:11.118Z)

```

* branch          master      -> FETCH_HEAD
3b36f9f docs(CLAUDE.md): truth-sync the stale Current-state section to shipped reality (audit
R2-5)
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
---merge-base---
e4101365cb5f3a608e76f8baf1d2e0d65ce1738e
e4101365cb5f3a608e76f8baf1d2e0d65ce1738e

```

---CLAUDE.md commits on master since merge-base---

---commits on branch---

3b36f9f docs(CLAUDE.md): truth-sync the stale Current-state section to shipped reality (audit R2-5)

9. user (2026-07-02T04:54:14.400Z)

backend/api/:

__init__.py

auth.py

job_worker.py

static

uploads.py

backend/compute/:

__init__.py

base.py

cloud_run.py

backend/storage/:

__init__.py

base.py

capabilities.py

gcs.py

gdrive.py

gdrive_api.py

local.py

ref.py

s3.py

---schema---

18:SCHEMA_VERSION = 9

107: # P2d upgrade-safety (no-op for a fresh DB or one already at SCHEMA_VERSION):

110: if version > SCHEMA_VERSION:

---min_crossings---

121: threshold is raised (`min\_crossings`/`min\_players` default 0 = OFF). The court mask

137: min_crossings: int = Field(default=0, ge=0)

10. assistant (2026-07-02T04:54:25.542Z)

Branch is cut from origin/master tip, single commit, CLAUDE.md-only diff. Now verifying the factual claims in the new text.

11. user (2026-07-02T04:54:29.587Z)

23:_KNOWN = ("local", "s3", "gcs", "gdrive", "gdrive-api")

54: A string with no ``scheme://`` prefix (including Windows ``C:\\...`` paths, which

61: scheme = m.group(1).lower()

63: if scheme == "gs":

64: scheme = "gcs"

65: if scheme in ("local", "gdrive", "gdrive-api"):

68: return StorageRef(scheme, "", rest, s)

69: if scheme in _KNOWN:

71: return StorageRef(scheme, bucket, key, s)

72: raise ValueError(f"unsupported storage URI scheme: {scheme}://{(in {s!r})}")

81: 1. An explicit ``scheme://`` on ``raw`` (`gs://`, `s3://`, `local://`,
`gdrive://`) ? as-is.

93: if _SCHEME_RE.match(s): # 1. explicit scheme wins

---compile-async---

479: """A human name for a video from its stored path/URI ? the filename stem

(``?/match\_2026.mp4`` ?

480: ``match\_2026``). Works for a local path, a bucket key, or a ``gdrive://`` URI (rsplit on
``/``).

541: NB: reels are stored FLAT in output_dir under the bare name passed to /api/compile ? the
engine

```
883:    ``/api/query`` + ``/api/compile`` read the rows with zero knowledge of where they were
produced.
1355:@app.post("/api/process", status_code=202)
1357:    """Submit a processing job (202 + job_id). Poll GET /api/jobs/{job_id} for state.
1426:        status_code=202,
1494:    """Stream a compiled highlight reel ? `filename` is the bare name given to /api/compile
1517:        detail=f"No compiled file named '{name}'. Compile first via /api/compile.",
1594:@app.post("/api/compile", status_code=202)
1595:def compile_highlights(req: CompileRequest, request: Request):
1596:    """Submit a reel-COMPILE job (202 + job_id); poll GET /api/jobs/{job_id} for the reel
(#329).
```

12. user (2026-07-02T04:54:31.943Z)

```
ABLATION_LAYER.md
ANALYZERS_AND_RECONCILERS.md
BACKLOG.md
CLOUD_GPU_DRYRUN_GUIDE.md
CODE_AUDIT_AND_TEST_HARDENING.md
CODE_CLEANUP_AUDIT_2026-06-24.md
CODE_MAP.md
COMMERCIALIZATION.md
COMPUTE_DECOUPLED_SERVING
CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md
CREATING_REELS.md
DATA_IN_GCS.md
DECISION_SNAPSHOT_REGRESSION.md
DEFERRED_HARDENING_PLAN.md
DESKTOP_APP_PLAN.md
DOC_STATUS.md
EVALUATION.md
EXECUTION_POLICY.md
EXPERIMENT_HARNESS.md
GOLDEN_REGRESSION_FIXTURES.md
HARDENING_LOOP_HANDOFF.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
I18N_PLAN.md
KHELSUTRA_PRODUCT_OVERVIEW.md
MULTI_SIGNAL_FUSION_PLAN.md
NEXT_STEPS.md
NIGHTLY_ENGINE_REGRESSION.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PACKAGING_PLAN.md
PERSONALIZATION_PLAN.md
PER_VIDEO_WORKSPACE.md
PLATFORM_ARCHITECTURE.md
POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
PROJECT_DOC_TEMPLATE.md
PROMINENT_COURT_DETECTION.md
QUALITY_ITERATIONS.md
R1_MILESTONE_LAUNCH_REPORT.md
R2_EVAL_METRIC_DESIGN.md
RALLY_DETECTION_CODE_REVIEW.md
---archives---
CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
COMMERCIAL_READINESS_REVIEW.md
DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
MONETIZATION_AUDIT.md
PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md
README.md
TEST_RUN_SUMMARY.md
checkpoints
decisions
```

13. user (2026-07-02T04:54:46.874Z)

README.md
TESTING_STRATEGY.md
architecture.svg
runlogs
1:docs/COMPUTE_DECOUPLED_SERVING/README.md:3:> **Status:** ? **M0 APPROVED** (owner, 2026-06-21)
? **IN PROGRESS** (build M1+) **. Owner:** Avi **. Created:** 2026-06-21 **. Last updated:**
2026-06-21
5:docs/COMPUTE_DECOUPLED_SERVING/README.md:41:| D14 | **NORTHSTAR** ? operate toward this: **a**
mobile, app-like flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the
stitched, **ready-to-share** reel lands **back in Drive**, next to the source**.** | owner
2026-06-21 | Implies a **`gdrive-api`** (OAuth) **StorageBackend** (read source + write reel back) ?
distinct from the local **mount** backend; **resolves S8 #7**, new **M12**. The mobile UI is the
khelsutra track; the engine must support it (async jobs + status + Drive round-trip / signed
URL). Net-new scope (OAuth + Drive write); not necessarily v1, but the design must not preclude
it (the **StorageBackend** ABC accommodates it). |
9:docs/COMPUTE_DECOUPLED_SERVING/README.md:113:| ID | Deliverable | Depends | Gated? | Status |
PR |
19:docs/COMPUTE_DECOUPLED_SERVING/README.md:152:- [] ? flip Status DONE, archive to
`docs/archives/past\_projects/`.
26:docs/COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md:3:> **Companion to**
[`README.md`](README.md) (\$7 tracker is the source of truth). **Status:** DRAFT (2026-06-21).

14. user (2026-07-02T04:54:47.917Z)

Next steps (live) ? refreshed 2026-06-21

> **How to read:** this top section is the **current cross-track state + prioritized pending**
list (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model**
& rally-quality track **plan** (2026-06-13) is **preserved below, still valid** ? it is now **one**
of several* active tracks. Live blow-by-blow: memory **current-state.md**. Doc lifecycle/hygiene:
docs/DOC_STATUS.md.

? Where we are (2026-06-21)

DECOUPLED_COMPUTE delivered on GCP + archived **(#270)**; **compute = GCP only** (ADR-013, DO
dropped). **COMPUTE_DECOUPLED_SERVING** subproject spun up (#272, **ADR-014**) ? one pure core
+ local/cloud profiles; **owner M0 design sign-off pending**. **Per-video workspace P0** shipped
(#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site +**
per-rally UI ship in parallel (sibling track). The **rally-quality / owned-model** track
(below) is the core ML work, gated on **growing the golden corpus**.

> **? LAUNCH DECISION ? NO-GO** (owner, 2026-06-30). The rally-detection product is **not**
cleared for an external/commercial launch; we stay in **development**. This is the expected
default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the **ship-gate**
target (#172) ? currently unset, needs corpus growth (and the R2 tolF1 ablation, item 8);
(2) resolve the **WASB shuttle-detector weight provenance** (commercial-clean licensing).
P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev PRs ?
launching. Re-decide only when both unblockers clear.

? Prioritized pending ? all tracks

P0 ? highest leverage / unblocks the most
0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE** ? "a few recordings
lying around ? highlights, from a page, no SSH". Today the cloud flow is **SSH/CLI-only**
(**docs/CLOUD_GPU_DRYRUN_GUIDE.md**); **M12** does NOT fix it. Needs (a separate khelsutra **web/`**
SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a **persistent** endpoint), **(ii)** the **SPA**
ingest/progress screen (khelsutra **web/`** B-P2: enter/upload video ? **/api/process`** ? poll
/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, **(iv)** a **compute PICKER** in the UI ("your machine vs our
cloud", D13/D15). The engine seams make it **possible**; the SPA makes it **usable**. Full record:

[[ui-driven-cloud-serving-gap]].

0b. ****?** [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING\_PLAN.md`. ****** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the ****bundled SPA**** single-origin, ****torch-free****, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). ****?** Next (fresh focused pushes): **** (1)**** P3 ****GPU validation + screencast**** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **** (2)**** P2b-install (uv/script + truly-local default config), **** (3)**** P2c first-run wizard (cross-repo khelsutra SPA from `api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

1. ? **** COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED**** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), ****M5 #286, M6 #287, M8 #288, M9-auth #290**** all merged (default-OFF); the ****real M7 L4 run**** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). ****Owner residual**** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a ****\$100/?10k cap**** (D17).

2. ****[owner, ongoing]** Grow the golden corpus ****?** multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track ****and**** full-corpus training feed on it.

3. ****[owner + Claude · GPU]** Full-corpus training (n?5) ****?** a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).

****P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)****

4. ****[Claude]** `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12 **** (M5/M6/M8/M9-auth ? merged this session). **M11**** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked `consent\_attested` (? privacy-sensitive ? give it fresh focus). ****M12**** = `backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to `\_KNOWN` + `\_backend\_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? ****M9-consent cols = a new v7 migration**** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.

5. ****[Claude]** Per-video workspace P1?P6 **** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` *[[the `output/chunks\_` de-hardcode base = ? done via #282]]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.**

6. ? ****Ops-Agent codified (#291)**** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

7. ****[owner]** Set the ship-gate target (#172) ****?** needs corpus growth.

8. ****[owner-side GPU]** R-series: promote-or-reject `R2` toLF1 as the gate **** (the ablation) ? needs GPU-extracted golden features.**

****P2 ? later / gated / deferred****

9. Rally-quality ****P3/P4**** (person-fusion LOVO, audio cue) ? needs GPU golden features.

9b. ? ****DONE (#396):**** the R4 density fix (finding B / P2b in `RALLY\_DETECTION\_FIXES\_PLAN.md`) is ****promoted to default-ON****. Golden-set A/B at the ****SERVED**** windowing showed toLF1 0.353?0.364 (?toLF1 +0.010, no regression), so `inactivity\_temporal\_density` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.

10. ****M0.4 scorer swap**** ? TemporalMaxer (per the 2026-06-17 external review).

11. ****Multisport**** (table-tennis first, then pickleball).

12. Serving ****M8/M9**** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE\_DECOUPLED\_SERVING`; owner/legal-gated).

13. ****WASB-C3**** weight provenance (gates `sharing` a trained model) ? legal.

14. ****Rename repo off "badminton"***** (owner, BACKLOG).

15. Owner housekeeping: retire the shared ****DO droplet**** + ****DO token****; collector ****`md5`**** export (collector repo).

16. ****PMF field validation**** (owner-side, business track).

****Doc-hygiene (tracked in `docs/DOC\_STATUS.md` §4):**** ? #1 per-video reconcile + ? #2 this refresh; ****pending:**** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary

confirm); confirm the COMPLETED top-level stubs; verify `HOSTING\_PLAN`/`UI\_PROPOSAL` currency.

Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

****Updated:**** 2026-06-13 (****rally-detection quality track****: fusion + experiment-harness design docs merged, owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the ****UI-alpha track opened****: ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`; ****PR-1 async jobs (#16) MERGED 2026-06-11, PR #87****; ****PR-2 upload/download MERGED 2026-06-11, PR #99**** ? live-E2E-verified (3-part chunked upload MD5 byte-identity + compile?browser-download); next: PR-3 identity. Same day: the ****8?9 quality sweep #90?#98 landed**** ? ruff/mypy/coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into `trajectory\_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
****Authoritative roadmap:****
`docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (+ ADR-008 for topology). ****Latest blow-by-blow:**** memory `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)

****M0 in-container quick wins are COMPLETE:**** ship-gate provisional blocker (****#170****), train/eval split guard (****#171****), the ****doc-consistency sweep**** (****#174**** ? corrected 16 drifted docs vs the code), and the ****P3 person-feature fusion litmus machinery is already present + suite-green**** (`backend/eval/fusion\_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a ****GitHub Actions minutes/billing outage**** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

****Immediate next step ? the real P3 LOVO measurement**** (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden features + `rallies.csv` labels are not yet here):

1. ****[owner / GPU box]**** `python -m backend.eval.fusion\_golden --manifest output/human\_lovo\_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `rallies.csv` labels are owner-side only.)
2. Artifacts land in the ****shared cloud folder**** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: ****[`GOLDEN\_DATA\_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)****; tracked in ****#175****.
3. ****[CPU dev/agent session]**** `python -m backend.eval.fusion\_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test\_ablation.py::test\_litmus\_reproduces\_gate\_a`).

15. assistant (2026-07-02T04:55:14.911Z)

NEXT_STEPS.md corroborates the NO-GO block, item 0, and gating claims. Now let me verify the cited commits, ADR-014, the Gate-A/fusion_hybrid wiring, and run the link check.

16. user (2026-07-02T04:55:19.455Z)

8bcf6ce feat(api): make /api/compile async (202 + poll) to kill the slow-stitch 524 (#329) (#339)
334ad0f feat(serving): M11 data-flywheel skeleton ? capture?shadow-eval?promote (default-OFF + consent faked-deny) (#336)
819e395 feat(serving): M12 ? gdrive-api:// OAuth/API StorageBackend (D14 Northstar) (#338)

6d3d054 feat(serving): SPA compute-picker ? per-request local/cloud override on /api/process (item 3a) (#335)
d92cc10 feat(serving): make the cloud compute-picker path work + provable (provenance + skip_ai_handoff forward) (#341)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted (\$0) (#267)
4a7bfbe docs: record the launch NO-GO decision (owner, 2026-06-30) (#404)
c8a493d fix(#398): served Gate-A litmus passes a typed IndexingConfig (not a dict) (#401)

17. user (2026-07-02T04:55:21.219Z)

637: (owner sequencing). Seed scope below. **? REALIZED in ADR-014 (2026-06-21):** the subproject is
639: (not just cloud serving); ADR-014 expands this 7-point seed into the M0?M10 plan.
686:## ADR-014: Compute-decoupled rally serving ? one pure core, profile-selected compute + storage
691:`ADR-005` (permissive licensing ? engine + image stay MIT/Apache/BSD) · `ADR-008` (train==serve featurizer single-sourcing ? the parity discipline this extends to local?cloud) · `ADR-009` (**local-first delivery** ? origin of the truly-local profile) · `ADR-010` (DECOUPLED DO?GCP) · `ADR-011` (**deferred M3 serving + M4 billing**; overrode "rent nothing / not a service"; \$06 gates = DPA/consent, collector `md5`, WASB-C3) · `ADR-012` (**Cloud Run+GPU = serving model**; NVDEC lever) · `ADR-013` (**GCP sole compute stack**) · **? ADR-014** (this). Non-ADR: realizes `PLATFORM\_ARCHITECTURE.md`'s `ComputeBackend`/`StorageBackend` registries; the truly-local profile = `VIDEO\_LOCALITY\_MODEL.md` L0 + `DESKTOP\_APP\_PLAN.md` `LocalDesktop`; informed by the archived `DEPLOY\_REPORT\_GCP\_2026-06-20` (cross-GPU sub-pixel parity finding).
---fusion_hybrid gate A---
8:- ``\_enrich\_candidate\_stats`` ? always adds the **net-crossing count** (Gate A signal,
16:- ``\_select\_for\_handoff`` ? optional served Gate A/B: when ``indexing.fusion.min\_crossings``
33:from backend.pipeline.detectors import rally_gate
42: """Shuttle trajectory (primary) + net-crossing Gate A + optional person cue (Gate B)."""
51: # Cache the play-axis/net estimate per trajectory so rally_gate doesn't re-estimate it
64: "net-crossing Gate A + optional person-occupancy Gate B adjudicate them, and the
"
125: # over the whole trajectory by rally_gate (camera-agnostic); reused across windows.
126: gated = rally_gate.gate_windows(
130: min_crossings=0,
151: """Cache rally_gate's (axis, net, span) play-axis estimate per trajectory so it's
156: self._axis_cache = rally_gate.estimate_play_axis(points)
205: """Served Gate A/B over the (already-enriched) per-window stats. Default thresholds
211: where no segment clears ``min\_crossings``), the gate falls back to the full ungated
212: set rather than emit a video with zero rallies. This makes Gate A/B strictly
219: min_crossings = int(fcfg.min_crossings)

18. user (2026-07-02T04:55:57.642Z)

docs/NEXT_STEPS.md:101:- **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ?
`backend/pipeline/detectors/rally\_gate.py` already implements the net-crossing ***"Gate A"*** (`apply\_gate(..., min\_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min\_crossings` knob in `IndexingConfig`).
Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win. (P2 FusionSegmenter specifics: registry `fusion\_hybrid`, **default-OFF**, seeds from substrate, persists `net\_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)
docs/NEXT_STEPS.md:104: 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)
---served gate a litmus---
backend/eval/served_gate_a.py
3b36f9f docs(CLAUDE.md): truth-sync the stale Current-state section to shipped reality (audit

```
R2-5)
c8a493d fix(#398): served Gate-A litmus passes a typed IndexingConfig (not a dict) (#401)
32f4ee6 test(eval): re-ground served Gate-A litmus + QUALITY_ITERATIONS on the 15-clip golden
corpus (#322)
e358b28 feat(eval): prominent-court window gate ? drop multicourt background-court rallies
(default-OFF) (#243)
a240221 fix(test): ablation litmus uses canonical experiment._is_fusion_schema (unbreaks the
suite) (#237)
---density default---
205:    inactivity_temporal_density: bool = True
```

19. user (2026-07-02T04:55:59.152Z)

[check-md-links] clean ? no broken internal links in live docs.
EXIT=0

20. user (2026-07-02T04:56:38.046Z)

```
3b36f9f1a268688c124f76f2c9733043cd8dd433 2026-07-02 10:19:58 +0530 Avi Dullu
---indexer endpoint---
1:# PEP 621 packaging for the badminton/sports highlight indexer (DEFERRED #7).
17:name = "badminton-highlight-indexer"
89:# `indexer --server`, `indexer --help`, etc. -> backend/main.py::main()
90:indexer = "backend.main:main"
---capabilities/assets endpoints (audit mentioned)---
429:@app.get("/api/capabilities")
727:@app.post("/api/assets", status_code=201)
936:    Surfaced in ``/api/capabilities`` so the SPA only offers the cloud toggle when it would
work."""
```

21. assistant (2026-07-02T04:57:13.581Z)

One last ground-truth check on the "neutral" wording ? the underlying litmus record.

22. user (2026-07-02T04:57:21.526Z)

```
1: ""SERVED Gate-A regression litmus ? the *production* FusionSegmenter path on real footage.
26:DROPS windows, on this under-segmented corpus it is mildly **F1-negative** (mean served ?F1 ?
27:-0.023), not the ~neutral -0.008 the original 6 showed. Under the permissive PROPOSAL
windowing the
28:same gate is ~F1-neutral (?F1 ? +0.03) while still cutting over-segmentation (1.36?0.79).
88:# ---- #398 regression: golden-free (runs in CI; only the golden DATA is absent there, not
cv2) ----
106:    assert isinstance(out, list) # no AttributeError from the dict-vs-typed-config
regression
128:def test_served_gate_a_does_not_materially_regress_f1():
129:    ""SERVED no-material-regression guard: on the production windowing Gate A must not
MATERIALLY
132:    windows and is mildly negative, NOT the ~neutral it was on the original single-court 6.
We pin
133:    a band that admits this measured mild loss but trips on a material regression.""
147:    (the direction-of-effect invariant ? even where the absolute lift is ~neutral).""
171:# p=0.031; archived). On the present multicourt corpus that lift is GONE ? the gate is
~F1-neutral
```

23. user (2026-07-02T04:58:06.466Z)

Structured output provided successfully